

Background: Duke Kunshan University (DKU) is an interdisciplinary institution that grants dual undergraduate degrees, an MOE Chinese degree and a degree from Duke University in Durham, United States. The principal structure of DKU majors is robustly interdisciplinary. No student confines their study to a single discipline (for example, biology or economics). Instead, all students engage in broad inquiry related to a subject or question (for example, political economy or global health) and take a wide variety of courses related to that area (for example, in public policy, history, ethics, or economics). As a result, our graduates are prepared to engage in a wide variety of inquiries using multiple methodologies to address complex issues that require interdisciplinary approaches.

This has implications for our vision and expectations of undergraduate theses and design projects, which reflect this broad interdisciplinary training. At DKU, every student completes a two-year project known as signature work which consists of multiple interconnected parts including thematic courses, experiential learning, capstones, and a final product. It seeks to integrate students' interdisciplinary educational experience and culminates in the creation of a product in a scholarly, creative, or applied nature in lieu of an undergraduate thesis or design required by JED. Because DKU encourages students to cultivate their independence and creativity as one of its institutional student learning outcomes, the student-led signature work projects often reflect students' own particular interdisciplinary interests and training. In addition, signature work has an intensive emphasis on problem-solving and skill-development which is much needed for any interdisciplinary inquiry; thus, students' final products are evidence of transferrable

skills that students have acquired and demonstrated through the 2-year program, rather than content knowledge narrowly defined by disciplinary training.

In sum, while the Chinese major declared with any given student might be construed narrowly, the experience of our students is much broader—and intentionally so. This is a distinctive feature of our curriculum, and this distinctiveness results in broadly interdisciplinary submissions from our graduates' submitting theses or design projects. We have designed this to prepare our students for a wide variety of graduate programs in China and the West, where interdisciplinary training is a competitive advantage.

Duke Kunshan University

Division of Social Sciences

Signature Work Thesis

**A QUANTITATIVE GREENWASHING INDEX FOR THE LUXURY SECTOR: A TEXT MINING
AND LDA APPROACH TO ASSESSING PREVALENCE OF GREENWASHING IN NEWS
MEDIA AND COMPARISON WITH COMPANIES' CSA SCORES**

by

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ABSTRACT

Greenwashing presents a structural barrier to genuine sustainability, particularly within the luxury fashion sector where brand equity is paramount. This research investigates the "Talk-Walk" gap by developing a quantitative Greenwashing Index (GW Index) to audit the relationship between corporate performance and public narrative. Utilizing a corpus of 21,150 XML documents extracted from ProQuest TDM Studio, the study employs a hierarchical entity resolution framework to map media scrutiny across 100+ subsidiary brands to their respective parent conglomerates.

The primary methodological innovation involves applying context-window Latent Dirichlet Allocation (LDA) to computationally filter thematic noise and isolate core critiques regarding transparency. The final analysis of 32 major luxury fashion brands reveals an important "Narrative Decoupling." Statistical results show insignificant Pearson and Spearman correlations, demonstrating that objective disclosure scores, such as the S&P Global CSA, have no predictive power over media-based greenwashing risk. This null result, further supported by a lack of discernible patterns in quadrant and slope visualizations, suggests that corporate sustainability ratings and public narratives function as entirely independent systems of evaluation. The study concludes that not enough data is available from institutional reporting frameworks to reflect the reputational dynamics of environmental communication, highlighting a critical need for narrative-based risk assessment in sustainability governance.

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Brand Name	ESG Score	CSA Score	G_Count	C_Count	Greenwashing Ratio
LVMH Moët Hennessy - Louis Vuitton, Société Européenne	63	62	1893	109879	0.017228042
Kering SA	78	78	830	42072	0.019728085
Compagnie Financière Richemont SA	37	32	370	36465	0.010146716
Burberry Group plc	55	45	174	15472	0.011246122
Hermès International Société en commandite par actions	65	64	96	12834	0.007480131
Capri Holdings Limited	39	40	296	26607	0.011124892
Tapestry Inc	40	44	848	36911	0.022974181
Prada S.P.A	43	51	174	11558	0.015054508
Moncler S.P.A	90	90	31	4292	0.00722274
PVH Corp.	42	36	51	4794	0.010638298
Puig Brands, S.A.	53	46	16	710	0.022535211
The Swatch Group AG	46	42	4	3116	0.001283697
OTB Group			927	35261	0.026289668
Chanel			761	19612	0.038802774
Rolex			39	5038	0.007741167
Salvatore Ferragamo S.p.A.	70	69	33	2405	0.013721414
Ralph Lauren Corporation	42	37	90	8642	0.010414256
Hugo Boss AG	85	84	9	5090	0.001768173
Armani			138	4470	0.030872483
Valentino			75	5706	0.013144059
Dolce & Gabbana			14	1176	0.011904762
Ermenegildo Zegna N.V.	59	58	35	4212	0.008309592
Guess?, Inc.	37	35	227	17170	0.013220734
PUMA SE	76	75	150	11927	0.012576507
Lacoste			13	976	0.013319672
Vans			271	15863	0.01708378
Patek Philippe			1	222	0.004504505
Audemars Piguet			6	223	0.02690583
Chopard			14	635	0.022047244
Breitling SA	46	45	4	214	0.018691589
Stella McCartney			280	13672	0.020479813
Jacquemus			1	830	0.001204819

Table 1. Shows ESG/CSA scores, Greenwashing/Climate counts and Greenwashing Ratio for all parent and other independent brands

Topic Number	Defining Keywords
Topic 1	<i>datacenterdynamics, he, ltd, power, our, center, energy, website, source, data</i>
Topic 2	<i>report, details, click, our, here, private, buysellsignals, reading, research, university</i>
Topic 3	<i>fair, release, press, 2022, source, website, tr, leather, university, td</i>
Topic 4	<i>2021, private, million, 11p, our, linklaters, sterling, shearman, tr, td</i>

Table 2. This table represents the raw thematic state of the corpus, where institutional "noise" and publisher boilerplate dominate the signal.

Topic Number	Defining Keywords
Topic 1	<i>greenwashing, public, business, allegations, news, materially, information, false, misleading, claims</i>
Topic 2	<i>statements, investigation, issued, ilm, investors, news, article, claims, false, misleading</i>
Topic 3	<i>vps, flaring, ovhcloud, tempers, private, misleading, claiming, customers, fair, leather</i>
Topic 4	<i>climate, critiques, effectiveness, concerns, transparency, regarding, challenges, initiatives, sustainability, greenwashing</i>

Table 3. This table represents the refined signal after the sentence-level context window and aggressive noise reduction were applied.

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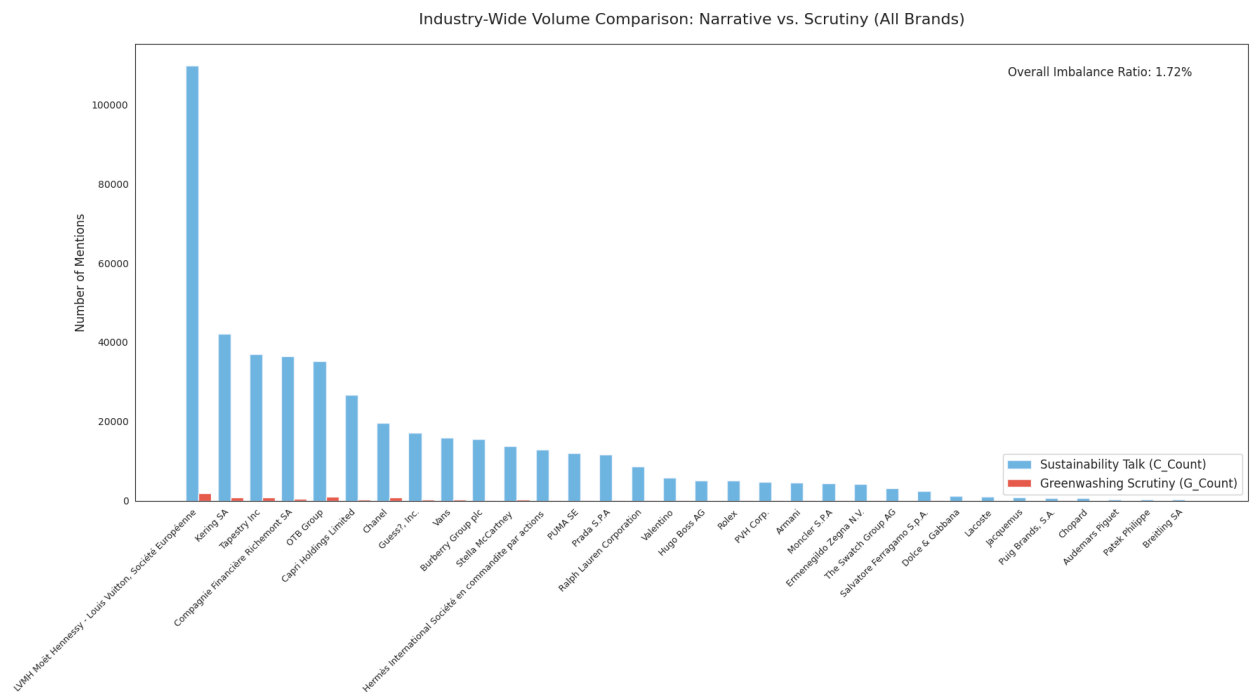


Figure 3. The Grouped Bar Chart showing C_Count vs G_Count for all 32 brands.

The "Talk-Walk" Gap Analysis

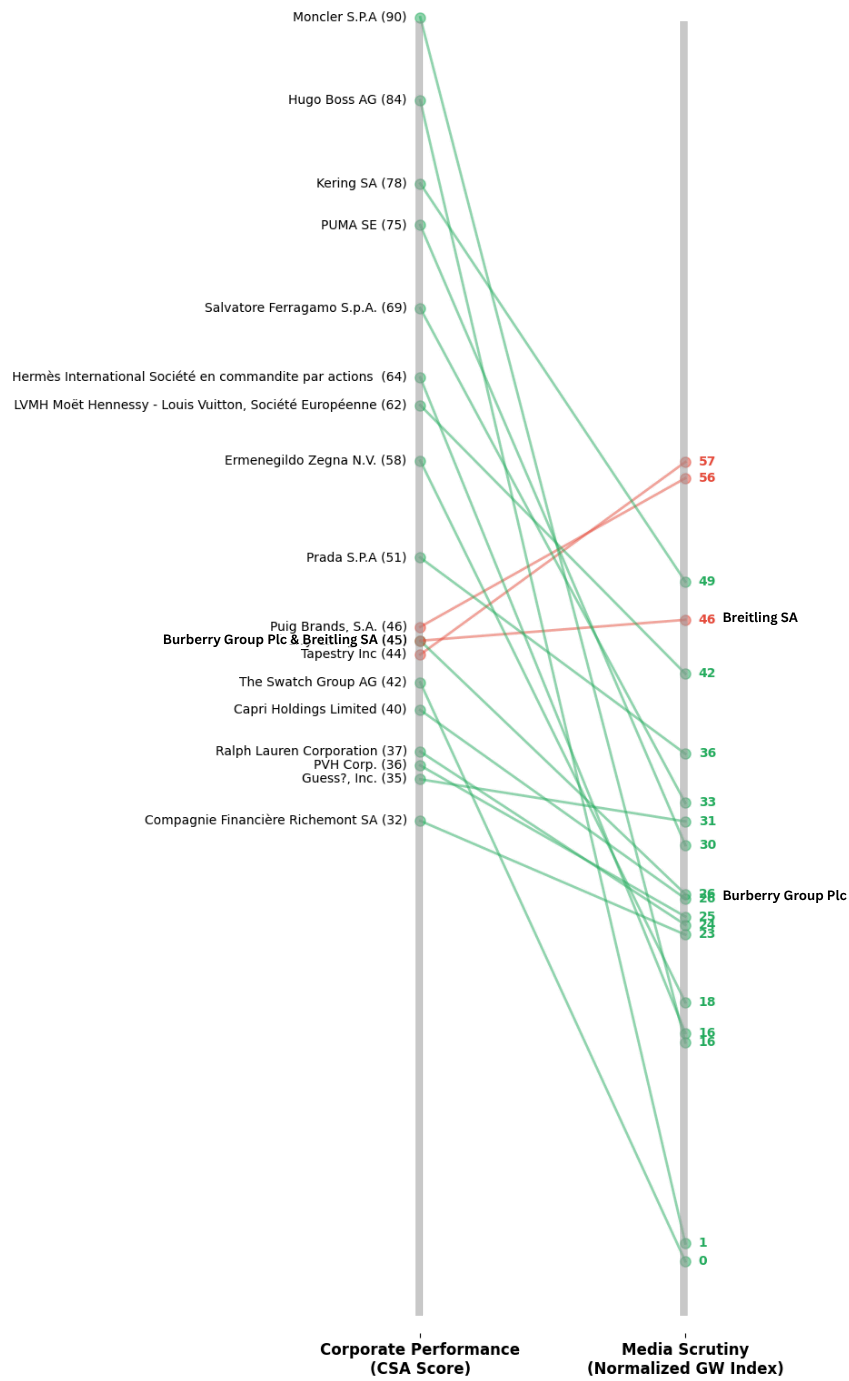


Figure 4. Slope Chart Revealing the Gap between CSA scores and Media Scrutiny (manually edited using canva for clarity).

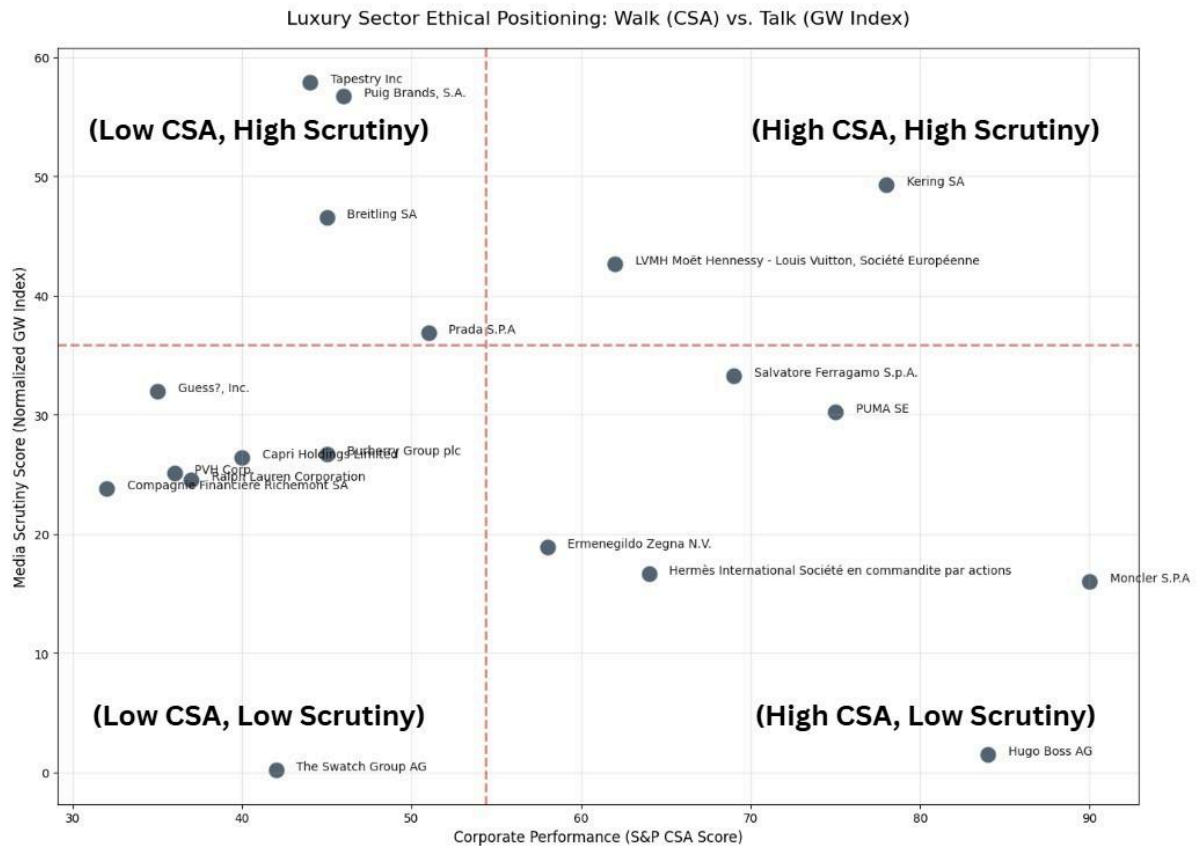


Figure 5. The 4-Quadrant Ethical Positioning Map with Mean Lines for 19 Luxury Brands with their CSA Scores (manually edited using canva for clarity).

KEYWORDS: Greenwashing, Luxury fashion, News Media, Corporate Sustainability Assessment (CSA), Latent Dirichlet Allocation (LDA)

INTRODUCTION

Corporate sustainability reporting has become a defining feature of contemporary environmental governance. Over the past two decades corporations across numerous industries have adopted Environmental Social Governance frameworks designed to measure environmental performance and communicate sustainability commitments to investors and consumers. These frameworks often produce numerical ratings that allow companies to benchmark their environmental performance relative to competitors.

Despite the rapid growth of sustainability reporting, critics have raised concerns that these frameworks do not necessarily capture genuine environmental progress. Companies may publicly promote sustainability initiatives while continuing environmentally harmful practices within their supply chains. This phenomenon is widely known as greenwashing.

Greenwashing is particularly relevant in the fashion industry, which is frequently criticized for its environmental footprint. Fashion production involves complex global supply chains that consume significant resources and generate large quantities of waste. At the same time, fashion brands increasingly advertise sustainable materials, ethical sourcing, and circular production systems in order to appeal to environmentally conscious consumers.

The central research question of this project is whether companies with stronger sustainability ratings are actually less likely to face accusations of greenwashing in media coverage. If CSA ratings accurately reflect environmental performance, companies with high scores should experience fewer accusations of misleading environmental communication. However, if CSA ratings primarily reflect corporate disclosures rather than actual practices, the relationship between ratings and public scrutiny may be weak.

To investigate this question, this research develops a computational method for analyzing large scale media coverage related to luxury fashion brands. By constructing a Greenwashing Index derived from news articles and comparing it with ESG scores, the study seeks to measure the gap between corporate environmental communication and public narrative.

LITERATURE REVIEW

The concept of greenwashing has been widely studied in environmental governance and marketing literature. Early research primarily focused on deceptive advertising practices in which companies exaggerated environmental claims to improve brand reputation. Over time, the scope of the concept has expanded to include corporate sustainability disclosures, environmental reporting practices, and broader forms of strategic communication that may misrepresent environmental performance.

A comprehensive review by de Freitas Netto et al. (2020) defines greenwashing as the misalignment between corporate environmental claims and actual environmental practices. Their systematic review identifies several recurring forms of greenwashing,

including vague environmental labeling, selective disclosure of favorable metrics, and misleading communication strategies that exploit regulatory gaps. These practices enable firms to present environmentally responsible images without implementing substantive operational changes.

Within the fashion industry, greenwashing has emerged as a particularly significant concern due to the sector's complex global supply chains and high environmental impact. Badhwar et al. (2024) describe greenwashing as a widespread phenomenon across the fashion value chain, ranging from misleading sustainability claims regarding raw material sourcing to deceptive marketing strategies targeting environmentally conscious consumers. Their systematic review highlights how such practices create consumer confusion and undermine trust in corporate sustainability initiatives.

Similarly, Adamkiewicz et al. (2022) document a variety of greenwashing tactics within the fashion industry. These include fake eco-labeling certifications, exaggerated claims regarding recycled materials, and symbolic sustainability campaigns that distract consumers from underlying environmental impacts. Their research emphasizes that greenwashing frequently occurs among companies that simultaneously maintain strong corporate social responsibility messaging.

Expanding on this critique, Qu (2024) argues that deceptive sustainability communication represents a structural barrier to meaningful environmental reform within the fashion sector. According to this perspective, misleading sustainability claims reinforce existing production models based on overconsumption and rapid product cycles, thereby delaying systemic change within the industry.

A central theoretical framework used to analyze greenwashing behavior is the concept of organizational decoupling. Decoupling occurs when firms publicly adopt sustainability commitments or governance policies while failing to implement corresponding environmental practices in operational reality. Research by Roszkowska-Menkes et al. (2024) demonstrates that companies often engage in selective disclosure strategies, emphasizing highly visible governance commitments while omitting information related to supply-chain practices or labor conditions, particularly following reputational controversies.

Empirical research examining firm-level greenwashing further highlights the importance of identifying discrepancies between communication and performance. In a systematic review of more than one hundred empirical studies, Lubloy et al. (2024) show that many greenwashing indicators rely on identifying inconsistencies between corporate sustainability disclosures and external performance metrics such as environmental ratings, emissions data, or independent sustainability assessments. Their analysis also emphasizes significant methodological limitations within the literature, noting that many studies rely on indirect proxies rather than observable evidence of misleading claims.

These challenges are particularly visible within the luxury fashion sector. A longitudinal analysis by Jestratijević, Rudd, and Uanhoro (2024) finds that luxury fashion brands frequently exhibit significantly lower supply-chain transparency compared to mass-market brands. Luxury firms often justify limited disclosure by citing the need to protect brand exclusivity and proprietary production processes. As a result, sustainability reporting in this segment can be particularly opaque.

Even when luxury companies provide sustainability metrics, the relationship between reporting and environmental impact may remain ambiguous. Karaosman, Brun, and Morales-Alonso (2015) analyze sustainability reports from luxury fashion supply chains and find that although many companies report numerous environmental key performance indicators, these disclosures often appear decoupled from the industry's relatively high carbon intensity and resource consumption.

These transparency challenges are also reflected in the relationship between greenwashing behavior and institutional Environmental, Social, and Governance (ESG) ratings. Islam (2024) analyzes European apparel companies and finds little statistical relationship between ESG scores and the presence of greenwashing behavior. These findings suggest that ESG frameworks, which frequently rely on self-reported disclosures, may not reliably identify misleading sustainability claims.

In response to these measurement limitations, researchers have increasingly turned to computational text analysis to study corporate environmental communication. Gourier and Mathurin (2024) developed a news-based Greenwashing Index using natural language processing techniques such as term frequency - inverse document frequency (TF-IDF) weighting and cosine similarity to analyze media coverage of corporate sustainability narratives. Their work demonstrates how large-scale textual datasets can be used to detect discrepancies between corporate claims and critical media scrutiny.

Similarly, Ardia et al. (2022) constructed a Media Climate Change Concerns index by analyzing climate-related news coverage. Their research demonstrates that media-based indices can capture shifts in public discourse surrounding environmental

issues and influence financial markets. Related approaches have been applied to corporate communications. Sautner et al. (2023) analyze earnings call transcripts using keyword-based natural language processing methods to measure firm-level exposure to climate-related discussions.

Environmental news coverage has also been used to study financial risk management. Engle et al. (2020) developed climate news indices to examine how environmental news shocks influence hedging behavior and investment strategies. Their research highlights the importance of measuring both the intensity and sentiment of climate-related media coverage when analyzing environmental risk exposure.

Recent advances in natural language processing have further expanded the toolkit available for detecting greenwashing behavior. Gorovaia and Makrominas (2025) apply NLP techniques to corporate social responsibility reports and find that firms associated with environmental controversies tend to produce longer, more optimistic, and less readable reports. These linguistic characteristics may function as attempts to obscure negative environmental information.

Building on these advances, Sharma et al. (2025) developed the “GreenShield” pipeline, which uses transformer-based language models to compare sustainability claims made in public advertisements with statements contained in corporate sustainability reports. By measuring semantic inconsistencies between these texts, the system can automatically flag potential instances of greenwashing.

Large-scale textual analysis also introduces methodological challenges related to keyword discovery and document retrieval. King, Lam, and Roberts (2017) demonstrate

that human researchers frequently struggle to identify the full range of relevant keywords required for comprehensive document searches. Their work proposes computer-assisted keyword discovery techniques to systematically identify relevant search terms within large textual datasets.

Finally, understanding consumer responses to sustainability communication is critical for interpreting the reputational consequences of greenwashing. Wang and Jung (2025) find that deceptive sustainability practices, such as false labeling or the omission of negative environmental information, significantly increase consumer perceptions of corporate hypocrisy, thereby reducing brand trust and purchase intentions.

Complementing this behavioral perspective, Jain (2018) provides a conceptual framework for sustainable luxury consumption, emphasizing that personal values, social identity, and cultural norms strongly influence consumer purchasing decisions. These factors may lead consumers to adopt sustainability-oriented consumption patterns without necessarily possessing the information required to verify corporate environmental claims.

Despite these advances, several gaps remain within the literature. Existing studies rarely examine the luxury fashion sector specifically in a longitudinal context, and few studies combine media-based measures of greenwashing with comparative ESG rating analysis. This research addresses these gaps by constructing a computational Greenwashing Index derived from large-scale media coverage and comparing it directly with firm-level sustainability ratings to assess the relationship between sustainability communication and external ESG evaluations.

METHODOLOGY

Research Design

This study adopts a communication imbalance framework for measuring corporate greenwashing risk within the luxury fashion industry. The analytical structure is derived from the “Talk versus Walk” framework proposed by Gourier and Mathurin (2024), which conceptualizes greenwashing as the divergence between symbolic sustainability discourse and verifiable environmental action. Rather than evaluating individual claims in isolation, this approach measures the broader informational environment surrounding firms to detect whether sustainability communication appears disproportionate relative to concrete climate-related activities.

Within this study, the framework is adapted to a large-scale textual analysis of media coverage. Two distinct forms of communication are operationalized: sustainability discourse (climate commitments, circular economy programs) and critical discourse (accusations of misleading marketing or deceptive claims). These are conceptualized as proxies for corporate sustainability “talk” and “walk,” respectively. Greenwashing risk emerges when the volume of promotional communication significantly outweighs the evidence of scrutiny or verification within public reporting.

Data Collection and Corpus Construction: ProQuest TDM Studio

The textual dataset for this study was constructed through an iterative refinement process using ProQuest TDM Studio (ProQuest, 2026). The objective of the data

collection process was to obtain a corpus of news articles that captured both sustainability communication and potential greenwashing accusations associated with major luxury fashion companies over time.

The initial stage of data collection involved constructing two pilot datasets designed to capture the two sides of the communication imbalance framework discussed in the research design. The first pilot dataset represented the numerator of the future Greenwashing Index and consisted of articles containing explicit references to greenwashing or misleading environmental claims. The second dataset represented the denominator and consisted of articles discussing sustainability initiatives or climate related actions.

The Greenwashing Numerator Dataset was collected using the following keyword query:

(greenwash* OR "false claim" OR "misleading claim" OR deceptive* OR unsubstantiated* OR "window dressing" OR "fake green")

AND

("LVMH" OR "Kering" OR "Hermes" OR "Hermès" OR "Richemont" OR "Prada" OR "Burberry" OR "Ralph Lauren" OR "Tapestry" OR "Capri Holdings" OR "Moncler")

This search was applied across a ten year time window and retrieved 1,255 news documents. These articles represented media coverage in which luxury fashion companies were discussed in the context of accusations of misleading environmental communication or questionable sustainability claims.

The second dataset was designed to capture reporting on sustainability initiatives undertaken by luxury companies. This dataset was constructed using the following keyword query:

(climate OR sustainability OR ESG OR "net zero" OR "carbon footprint" OR "circular economy")

AND

("LVMH" OR "Kering" OR "Hermes" OR "Hermès" OR "Richemont" OR "Prada" OR "Burberry" OR "Ralph Lauren" OR "Tapestry" OR "Capri Holdings" OR "Moncler")

Applying this query across the same ten year period retrieved 13,653 documents. These articles represent reporting on climate strategies, sustainability initiatives, or environmental governance programs associated with the selected luxury firms.

At this stage the two pilot datasets appeared sufficient for constructing a greenwashing ratio. However a manual inspection of a sample of articles revealed a substantial irrelevancy problem. Although the search queries contained targeted environmental keywords, many articles mentioned the luxury companies only incidentally. For example numerous retrieved documents consisted of omnibus news summaries or financial market reports in which brand names appeared alongside unrelated discussions of geopolitical conflicts or macroeconomic events. Coverage of the Russia Ukraine war appeared particularly frequently because luxury companies were occasionally referenced in discussions of sanctions, supply chain disruptions, or global retail markets.

As a result only approximately one out of every twenty to thirty retrieved documents contained substantive discussion of sustainability practices or greenwashing controversies within the luxury fashion sector. The remainder contained incidental mentions that did not meaningfully contribute to the analysis of corporate environmental communication.

To address this noise a more restrictive search strategy was developed. The final dataset was constructed using a proximity based query that required luxury brand names to appear within close textual distance of sustainability or greenwashing related terminology. Proximity operators restrict the maximum number of words that can appear between two search terms and therefore ensure that the terms are discussed within the same semantic context.

The final query was structured as follows:

**("LVMH" OR "Kering" OR ... [80+ Brands] ... OR ("Hermes" NEAR/5 "luxury")
OR ...) NEAR/25 ("sustainability" OR "climate change" OR "emissions" OR
"greenwash*" OR "green-wash*" OR "misleading claim*" OR
"unsubstantiated" OR "false claim*" OR "greenhushing" OR "green-lighting"
OR "vague environmental" OR "deceptive marketing")**

Applying this refined query produced a high density corpus of 21,150 XML documents. Compared with the initial pilot datasets this approach significantly reduced irrelevant news coverage because the query ensured that sustainability or greenwashing terminology appeared in close proximity to luxury brand names.

A manual review of article metadata confirmed that the majority of retrieved documents contained substantive discussion of luxury fashion companies in relation to sustainability strategies, environmental controversies, or greenwashing accusations. Although some noise remained, the proximity based search dramatically increased the proportion of relevant articles within the dataset.

The combined corpus was then programmatically separated into two analytical categories representing Greenwashing Counts and Climate or Sustainability Counts. This classification was implemented using Python within a Jupyter Notebook environment. Keyword matching was used to identify whether an article primarily discussed misleading environmental claims or sustainability initiatives. Articles containing terms associated with accusations such as greenwashing, deceptive marketing, or unsubstantiated environmental claims were classified within the Greenwashing Count dataset. Articles discussing climate strategies, emissions targets, or sustainability initiatives were classified within the Climate Count dataset.

This approach allowed the study to preserve the contextual richness of the combined corpus while still maintaining the conceptual structure required for calculating a greenwashing ratio.

Data Collection and Corpus Construction: GDELT and NewsAPI

In an effort to identify a more accessible alternative to the complex XML extraction process of ProQuest TDM Studio, the study also explored the Global Database of Events, Language, and Tone (GDELT) and NewsAPI. These sources were evaluated for their ability to provide high-frequency, real-time data on greenwashing accusations

within the luxury sector. However, both platforms presented significant technical and thematic limitations that ultimately necessitated their categorization as secondary or "pilot" sources rather than primary datasets.

GDELT and Google BigQuery: Technical Constraints and Quota Management

The GDELT Global Knowledge Graph (GKG) was accessed via Google BigQuery to perform large-scale SQL filtering of environmental themes. This process was immediately met with extreme technical hurdles related to data quotas. Google Cloud imposes a 1 TB monthly limit on free-tier data processing; given the petabyte-scale of GDELT, initial broad queries depleted this quota within seconds. To circumvent these restrictions, the data collection required the use of seven distinct Google accounts to distribute the processing load. Furthermore, the query timeframe had to be split into two separate segments (2016-2019 and 2020-2025) to bypass session-level memory limits (Appendix A: Figure 3)

Two distinct SQL strategies were deployed: an initial pilot query targeting ten primary parent brands and a comprehensive "Parent-Child" resolution query targeting over 80 entities. Despite these engineering efforts, the results from GDELT revealed a critical lack of "signal." While the platform successfully retrieved a high volume of general sustainability discourse for the initial query of 10 brands, it failed to isolate specific, verifiable greenwashing accusations for the target luxury brands for both queries. Additionally, no data was retrieved for sustainability disclosure either for the comprehensive query. The results showed that while luxury companies are frequently mentioned in the context of environmental themes or general sustainability talk over

time (Appendix A: Figures 1 and 2), the specific GDELT theme for greenwashing yielded no data points to conduct a statistically significant correlation analysis.

NewsAPI: Data Scarcity and Content Irrelevancy

Parallel to the GDELT exploration, NewsAPI was utilized to query the same luxury brand list. However, the only document retrieved from it was a visual advertisement for Ralph Lauren that incidentally appeared alongside an unrelated article on textile technology, rather than a substantive report on environmental communication risk.

Ultimately, the failure of GDELT and NewsAPI to provide a granular greenwashing signal served as a critical methodological turning point. It proved that while general sustainability "talk" is pervasive across the open web, specific greenwashing "scrutiny" is a much rarer and more localized narrative. This justified the decision to rely exclusively on the deep, trade-focused archives of ProQuest TDM Studio, which provided the necessary linguistic density required to construct the final Greenwashing Index.

Development of the Text Processing Pipeline

A custom Python processing pipeline was developed to convert the raw XML corpus into a structured dataframe. Using the BeautifulSoup and lxml libraries (for working with XML documents), the script programmatically isolated the title, abstract, and body text of each document, concatenating them into a single string for analysis.

Preprocessing involved iterative refinement to address inconsistencies in the XML structure, such as publisher formatting artifacts and irregular tag structures. These

issues were resolved through the removal of HTML markup, normalization of whitespace, and the filtering of boilerplate text.

The development of this pipeline was supported by an iterative technical dialogue with the Gemini large language model. This interaction functioned as a methodological development environment, providing guidance on package optimization, proximity-based keyword searches, and troubleshooting for large-scale data iteration.

Hierarchy Entity Resolution Framework

A significant methodological challenge in the luxury sector arises from the discrepancy between brand-level media mentions (e.g., Gucci, Dior) and firm-level sustainability reporting (e.g., Kering, LVMH). To reconcile this difference, I implemented a hierarchical entity resolution framework that systematically mapped brand references to their respective parent companies.

I constructed a comprehensive lookup table in the form of a Python dictionary that mapped more than 100 individual luxury brands to 32 parent conglomerates and independent fashion houses. Each dictionary entry contained a parent company as the key and a list of associated brand aliases as the values.

To execute the entity resolution process, I wrote a Python program that iterated through each document in the textual corpus. For every document, I performed a case-insensitive substring search using Python's regular expression module. Rather than relying on a simple plaintext search such as grep, the program iterated through the list of brand aliases associated with each parent entity in the dictionary.

When a brand mention appeared within the document text, I programmatically attributed the document's sustainability and greenwashing counts to the corresponding parent company's aggregate total. This ensured that discourse surrounding subsidiary brands contributed to the reputational and communication profile of the corporate group that ultimately controls the brand.

This alignment process was essential because luxury conglomerates frequently communicate sustainability commitments at the group level, while media coverage often refers to individual brands. By consolidating subsidiary mentions into their respective parent entities, I ensured that the resulting Greenwashing Index accurately reflected the overall sustainability discourse surrounding each corporate group, enabling direct comparison with firm-level ESG and Corporate Sustainability Assessment scores.

Construction of the Greenwashing Index

The Greenwashing Index used in this study is conceptually derived from the index proposed by Gourier and Mathurin (2024), which measures discrepancies between corporate environmental communication and observable environmental performance indicators. In the context of this research the index is adapted to a textual measurement of discourse imbalance within media coverage.

For each company two counts are calculated from the dataset:

The first variable represents the number of articles in which the company is associated with accusations of misleading environmental claims. This value is referred to as the Greenwashing Count.

The second variable represents the number of articles discussing sustainability initiatives, climate strategies, or environmental governance activities associated with the company. This value is referred to as the Climate or Sustainability Count.

The Greenwashing Index for each brand was calculated using the following ratio:

$$GW_Ratio_{brand} = \frac{\sum G_Count_{brand}}{\sum C_Count_{brand} + 0.0001}$$

A small constant ($\epsilon = 0.0001$) was added to the denominator to prevent division by zero, ensuring a valid index value for all 32 analyzed entities (Table 1).

This ratio serves as a normalized indicator of the intensity of greenwashing related discourse relative to sustainability communication. Companies that receive extensive sustainability coverage but very few accusations of misleading claims will generate lower index values. Conversely companies that are frequently associated with greenwashing accusations relative to their sustainability coverage will produce higher index values.

The conceptual validity of the index is supported by existing literature on greenwashing measurement. Prior research has shown that discrepancies between sustainability communication and environmental performance often manifest through media scrutiny and investigative reporting. The index therefore captures the reputational dimension of greenwashing risk by measuring how often corporate environmental messaging is challenged within the public discourse.

In addition, the dataset itself was manually inspected to verify that articles classified within the greenwashing category genuinely contained discussions of misleading environmental communication. A random sample of documents was reviewed to confirm that the classification procedure captured meaningful critiques rather than incidental mentions of sustainability terminology.

The index was compared with external ESG performance indicators obtained from the S&P Global Corporate Sustainability Assessment. If the index captures meaningful reputational risk associated with sustainability communication, companies with weaker environmental performance should theoretically experience greater scrutiny regarding greenwashing claims. Correlation analysis between the index and ESG scores therefore serves as an indirect validation of the construct.

Although the index does not directly measure the truthfulness of corporate claims, it provides an empirically observable indicator of the degree to which sustainability communication is accompanied by critical media scrutiny.

Min-Max Normalization

Because the raw Greenwashing Ratios resulted in microscopic decimal values, a Min-Max Normalization was applied to transform the data into a standardized 0–100 scale. This process was essential for conducting a direct visual and statistical comparison with S&P Global's CSA and ESG scores, which operate on the same 100-point scale:

$$Normalized_GW = \frac{x - x_{min}}{x_{max} - x_{min}} \times 100$$

Topic Modeling and Noise Reduction

Since the dataset consisted of more than twenty thousand news articles retrieved from a wide range of international publications, additional analytical steps were required to ensure that the corpus actually reflected meaningful discourse surrounding sustainability communication and greenwashing accusations within the luxury fashion industry. Keyword filtering alone does not guarantee thematic relevance. Articles retrieved through keyword searches often contain publisher boilerplate, syndicated news digests, or unrelated content in which brand names and environmental terms appear coincidentally within the same document. As a result, a validation procedure was implemented using Latent Dirichlet Allocation topic modeling.

Latent Dirichlet Allocation is an unsupervised probabilistic modeling technique that identifies clusters of words that frequently co-occur within a corpus. These clusters, referred to as topics, represent latent thematic structures within the text. In the context of this research, the objective of applying LDA was not to produce definitive thematic interpretations but rather to verify whether the dataset contained coherent patterns of discourse associated with sustainability communication and accusations of greenwashing. Importantly, document-topic proportions derived from LDA were not used as inputs in the Greenwashing Index calculation. Instead, the model served as a diagnostic method to evaluate the signal-to-noise ratio of the dataset and to verify that the corpus meaningfully captured discussions of sustainability communication and greenwashing scrutiny within the luxury sector.

The initial LDA model was applied to the full corpus of raw XML documents. The results indicated that the dataset contained substantial publisher boilerplate and incidental mentions of brand names. As illustrated in Table 1, several of the dominant topics were characterized by technical and administrative terms such as “*datacenterdynamics*,” “*private*,” “*source*,” and “*university*.” These terms suggested that many retrieved documents consisted of aggregated news digests or technical publications where luxury brands appeared only tangentially. This outcome functioned as a diagnostic indicator of dataset noise and confirmed the need for further contextual refinement.

In response to this finding, the text-processing pipeline was adjusted to implement a sentence-level context window and filtered out all of the noise and irrelevant keywords by creating a separate dictionary to make sure they are not included as recurring topics. This process was repeated multiple times with each time more noisy words and irrelevant words being found were added into the dictionary until the LDA eventually retrieved topics which were a lot more relevant to this research.

This thematic shift confirmed that the refined corpus successfully captured the type of discourse required for the study. In particular, the presence of topics focused on transparency gaps, criticism of corporate claims, and allegations of misleading sustainability communication provided empirical evidence that the dataset contained the narrative structures underlying greenwashing scrutiny.

Therefore, while the LDA analysis was not used as a quantitative component of the Greenwashing Index, it played an important role in validating the dataset. The results demonstrate that the corpus contains meaningful discussions about the discrepancy

between corporate sustainability messaging and measurable environmental performance, which forms the conceptual basis of the “Talk–Walk” framework used in this research.

Data Visualization

Following the construction of the Greenwashing Index and the integration of external sustainability performance metrics, several visualization techniques were used to explore the relationship between corporate sustainability communication and media scrutiny. These visualizations were not used to generate the statistical findings themselves but rather to illustrate the structural patterns present within the data. Three visualizations were produced using Python data visualization libraries.

The first visualization is a grouped bar chart (Figure 3) comparing the total number of sustainability related articles and greenwashing related articles associated with each brand. This chart is designed to illustrate the relative magnitude of sustainability discourse compared with critical scrutiny across the entire dataset. For each company two counts are plotted: the Climate or Sustainability Count representing the volume of sustainability related coverage, and the Greenwashing Count representing the volume of articles containing accusations of misleading environmental claims. By presenting both values side by side for each brand, the chart provides an overview of the relative balance between promotional sustainability narratives and critical reporting.

The second visualization is a Slope Chart (Figure 4) comparing corporate sustainability performance scores with the normalized Greenwashing Index derived from media coverage. Each brand is represented by a line connecting two points. The points on the

left represent the firm's Corporate Sustainability Assessment score obtained from S&P Global. The points on the right represent the firm's normalized Greenwashing Index value calculated from the textual dataset. The slope of the connecting line visually illustrates the relationship between internal sustainability performance metrics and external media scrutiny.

The third visualization consists of a four quadrant positioning map (Figure 5) designed to illustrate how brands are distributed across two dimensions simultaneously. The horizontal axis represented corporate sustainability performance measured through CSA scores, while the vertical axis represented media scrutiny measured through the normalized Greenwashing Index. Mean values for both variables were used to divide the chart into four quadrants. Each brand was plotted within this space according to its relative position on the two dimensions.

These visualizations serve as exploratory analytical tools that helped illustrate the broader structure of the dataset and the potential relationship between sustainability performance metrics and greenwashing discourse in media coverage. Detailed interpretation of the patterns observed in these figures is presented in the Results section of this paper.

Statistical Analysis and Correlation

To evaluate the relationship between corporate performance and media scrutiny, the normalized GW Index was correlated with external ESG and CSA scores. Both Pearson (linear) and Spearman (rank) correlation analyses were performed to detect whether higher disclosure performance successfully mitigated greenwashing risk.

RESULTS

The analysis produced several important findings regarding the relationship between corporate sustainability scores and media narratives about greenwashing. The results are categorized by the thematic evolution of the corpus, the quantitative volume of discourse, and the statistical correlation between internal performance and external scrutiny.

The Latent Dirichlet Allocation (LDA) analysis provided empirical validation that the textual corpus contained substantive discourse related to sustainability communication and greenwashing scrutiny within the luxury fashion sector. Importantly, the LDA results were not used quantitatively in the construction of the Greenwashing Index; instead, they served as a diagnostic step to verify that the dataset captured the relevant narratives required for the study's "Talk–Walk" framework.

The first LDA model, estimated on the full XML document corpus, revealed that the dataset initially contained a substantial amount of non-substantive content. Several of the dominant topics in Table 1 are characterized by technical and administrative vocabulary, including terms such as *"datacenterdynamics," "private," "source,"* and *"university."* These results indicated that many retrieved documents were aggregated news digests or technical publications in which luxury brand names appeared only incidentally. The presence of these topics confirmed that the raw corpus contained significant background noise, reinforcing the need for further contextual filtering.

After implementing a sentence-level context window to isolate passages where brand names appeared near sustainability or greenwashing terminology, a second LDA model

was estimated on the refined text segments. The resulting topics exhibited substantially higher thematic coherence and were directly aligned with the conceptual focus of the study.

Topics 1 and 2 in Table 2 capture the regulatory and reputational risks associated with environmental claims, with prominent keywords including *“allegations,” “materially,” “false,”* and *“misleading.”* These terms reflect journalistic and legal discourse surrounding accusations of deceptive sustainability communication. Topic 4 contained terms such as *“transparency,” “effectiveness,” “critiques,”* and *“initiatives,”* which correspond to debates about the credibility and measurable impact of corporate sustainability strategies.

Together, these topics indicate that the refined corpus captures a discourse environment centered on the credibility of corporate sustainability messaging and the potential gap between environmental commitments and observable outcomes. This thematic structure aligns closely with the theoretical premise of the study, which conceptualizes greenwashing as a divergence between sustainability “talk” and measurable environmental “walk.”

The LDA results therefore function as empirical validation of the dataset, demonstrating that the textual corpus contains meaningful discussions of sustainability claims, corporate transparency, and allegations of misleading environmental communication in the luxury fashion industry. This validation supports the use of the corpus as the foundation for constructing the Greenwashing Index and analyzing sustainability discourse at the firm level.

The quantitative distribution of articles revealed a massive disparity between sustainability promotion and critical scrutiny. The industry-wide volume comparison for all 32 brands (Figure 3) illustrates that sustainability "talk" (*C_Count*) vastly outweighs greenwashing "scrutiny" (*G_Count*).

The most striking finding emerged from the statistical comparison between external sustainability ratings and the media-derived Greenwashing Index. To determine if higher corporate performance (S&P Global CSA scores) leads to lower media scrutiny, a dual correlation analysis was performed:

- Pearson Correlation (Linear): $r = -0.166, p = 0.497$
- Spearman Rank Correlation: $\rho = -0.026, p = 0.915$

Both tests returned near-zero coefficients and p -values significantly higher than the 0.05 threshold for significance. These results provide empirical evidence of "Narrative Decoupling." There is no statistically meaningful relationship between a brand's disclosure performance and its exposure to greenwashing accusations. A company with a high CSA score, such as Hugo Boss (84), Kering (78), or PUMA (75), is not consistently more protected from scrutiny than a company with a lower score.

The lack of statistical pattern is further visualized through the Slope Chart (Figure 4) and the Quadrant Map (Figure 5). In the Slope Chart, the lines connecting Corporate Performance (CSA) to Media Scrutiny (Normalized GW Index) cross haphazardly, with most companies, whether with a moderate or high CSA, having similar Normalized GW Index values. High-performing brands like Kering and PUMA show significant vertical

"gaps" between their internal scores and external scrutiny levels, indicating that media perception does not follow a linear path of performance.

The Quadrant Map reinforces this "randomness" with 19 luxury brands plotted across four ethical positions. The mean lines divide the sector into groups such as high-performers facing high scrutiny and low-performers evading it entirely. The broad spread of dots across all four quadrants, rather than a diagonal line, visually confirms that public narratives regarding greenwashing operate independently of the reporting frameworks used by institutional rating agencies.

DISCUSSION

The core objective of this study was to evaluate whether established corporate sustainability performance metrics could serve as a predictive shield against reputational greenwashing risk. The resulting statistical evidence, specifically a Pearson correlation of $r = -0.166$ and a Spearman rank correlation of $\rho = -0.026$ ($p > 0.05$), reveals a profound "Narrative Decoupling." This finding suggests that the luxury industry's sustainability landscape is characterized by an institutional tension: while corporate reporting frameworks (CSA/ESG) have become highly sophisticated and standardized, they appear to operate in a vacuum, isolated from the reputational dynamics of the global media narrative.

The absence of a meaningful relationship between these variables implies that institutional ratings primarily capture a firm's capacity for structured, voluntary disclosure rather than its actual immunity to public scrutiny. Media coverage frequently responds to volatile external triggers, investigative journalism, sudden regulatory shifts, or NGO

advocacy, that exist outside the rigid periodic cycles of ESG reporting. Consequently, even a company with an industry-leading CSA score can face intense greenwashing accusations if a specific narrative "mismatch" is detected by the public. This suggests that for luxury brands, transparency is a double-edged sword; high disclosure volume may inadvertently provide the media with a larger "surface area" for critique, leading to the erratic patterns observed in the Quadrant Map.

In the current informational environment, the data reveals a profound Narrative Asymmetry within the media ecosystem itself. The finding that sustainability "Talk" (*C_Count*) vastly outweighs greenwashing "Scrutiny" (*G_Count*) suggests that the media landscape functions more as a megaphone for corporate sustainability narratives than as an independent auditor of them.

Rather than the media being "drowned out" by external forces, these results indicate that the media ecosystem is self-saturating. By prioritizing high-volume, press-release-driven reporting on climate commitments and ESG milestones, the media creates a dense "noise" of promotional communication that effectively submerges its own investigative or critical "signal." This internal imbalance suggests that the greenwashing narrative is still in a nascent, reactive stage, struggling to find space within a media cycle that is already heavily committed to positive sustainability coverage.

This leads to a critical governance implication regarding "Narrative Buffering." If the media landscape is structurally biased toward high-volume promotional reporting, businesses can achieve a state of reputational safety simply through the sheer quantity

of their "green" output. By flooding the media arena with positive updates, firms may facilitate a form of "informational hegemony" where occasional scandals or policy failures are statistically overwhelmed by the media's own broad sustainability discourse before consumer attention can fully coalesce.

From a consumer perspective, this research highlights the psychological complexities of "Green Luxury." As identified by Jain (2018), luxury purchasing behavior is often driven by deeply ingrained social and personal values such as achievement, power, and security. Consumers may adhere to sustainability "norms" as a trend or status symbol without the technical capacity to audit the claims they consume. This research, therefore, serves as a necessary intervention, providing a framework for customers and external organizations to move beyond brand-managed narratives and demand more rigorous, independent verification.

LIMITATIONS

Despite the development of a novel quantitative index, several structural and data-related limitations must be acknowledged:

1. The primary limitation is the lack of a statistically significant association between GW Ratios and ESG/CSA scores. While the "no correlation" finding is theoretically valuable, the low *N* value (32 brands) and the sparse frequency of greenwashing mentions relative to climate talk mean that these results currently represent independent mentions rather than a definitive global pattern.
2. The study focused predominantly on luxury brands headquartered in Europe and the West. This Western-centric sample fails to represent the broader fashion

industry or the specific sustainability challenges faced in the Global South, where production impacts are most acute but media scrutiny may be less documented in the ProQuest TDM corpus.

3. Large luxury conglomerates possess the economic power to influence media narratives. The risk of "media pumping", where companies increase promotional output following a scandal to suppress negative coverage, makes it difficult to determine if a low GW Index reflects genuine ethical behavior or a highly successful PR strategy.
4. Proving greenwashing remains inherently difficult for private entities that do not release CSA data. Without internal audits or comprehensive Life Cycle Assessments (LCAs), a media-only index provides a window into *perception*, but not necessarily the *material truth* of a brand's environmental footprint.

CONCLUSION

This research successfully established a novel computational method for measuring the gap between corporate environmental communication and media scrutiny in the luxury sector. By analyzing over 21,000 documents and mapping the "Talk-Walk" dynamics of 32 major entities, the study demonstrates that formal sustainability ratings are not currently reliable predictors of greenwashing risk. Reputational dynamics operate through independent mechanisms that extend far beyond the formal reporting frameworks submitted by corporations.

The findings suggest that sustainability governance must evolve to incorporate external narrative analysis alongside traditional disclosure metrics. For governments and

pressure groups, this research highlights a need for stricter transparency and the realization that the current media landscape may not be a sufficient auditor of corporate claims. For consumers, the GW Index provides a tool to begin questioning the "halo effect" of luxury branding.

Future research should focus on expanding the sample size to include a more diverse global array of fashion brands and incorporating social media discourse to capture real-time consumer backlash. Additionally, integrating the GW Index with material impact data, such as Life Cycle Assessments, would bridge the gap between media perception and physical environmental reality. As sustainability communication continues to saturate the market, the ability to computationally deconstruct these narratives will become an essential requirement for true corporate accountability.

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SW NARRATIVE

The development of this Signature Work project represents the culmination of several interdisciplinary themes explored throughout my undergraduate education. The project integrates environmental governance research, computational social science methods, and data science techniques to investigate the relationship between corporate sustainability disclosures and public scrutiny in the luxury fashion industry. Through this work, I sought to address a central challenge in sustainability research: how to measure the gap between what companies claim and how their environmental performance is perceived externally.

The initial inspiration for this project emerged from my coursework on sustainability governance and environmental policy. These courses repeatedly highlighted the limitations of Environmental Social Governance metrics. Many institutional rating

systems depend heavily on voluntary disclosures submitted by corporations themselves. While such disclosures can provide useful information about internal policies and reporting structures, they often fail to capture real time reputational risks that arise when companies are accused of misleading environmental communication. This discrepancy led me to explore the phenomenon commonly referred to as greenwashing, which describes situations where companies exaggerate or misrepresent their environmental commitments in order to strengthen their public image.

In the early stages of this project, my research objective was relatively simple. I wanted to examine whether companies with strong sustainability scores were less likely to face accusations of greenwashing. However, as the research progressed it became clear that answering this question required developing a completely new dataset and analytical framework. Most existing greenwashing studies rely on qualitative case studies or survey research. Very few studies attempt to measure greenwashing accusations at scale using computational techniques. My Signature Work therefore evolved into a data driven investigation that combined text mining methods with sustainability metrics.

The technical foundation of the project was built through an extended process of experimentation and methodological revision. At the beginning of the research process I attempted to collect media data using publicly available global news databases. The first dataset I attempted to use was the Global Database of Events Language and Tone. This dataset is widely used in computational journalism and political science research because it contains millions of news articles and event records from international media sources. I attempted to query the dataset through Google BigQuery in order to retrieve

news articles mentioning major luxury fashion brands. However, this approach quickly proved impractical because the size of the database triggered strict quota limits on the query system. Each attempt to retrieve articles returned only a fraction of the required results before the system stopped execution due to computational limits.

Following this obstacle, I attempted a second strategy using the GDELT document application programming interface. This tool provides a simplified method for retrieving individual news articles related to specific keywords. Unfortunately, the interface only retains approximately ninety days of indexed content. Because my research required analyzing long term patterns of environmental discourse, the limited time horizon of the database made it impossible to conduct a meaningful analysis. A third attempt using the NewsAPI service also produced disappointing results. Even after constructing extensive search queries for luxury brands, the service returned only a handful of articles. These repeated failures forced a major pivot in the research design.

The project ultimately shifted to ProQuest TDM Studio, an academic platform designed for large scale text mining. Unlike conventional news databases, TDM Studio allows researchers to download thousands of full text documents directly for computational analysis. Using a structured search query containing sustainability related keywords and brand names, I extracted more than six thousand news articles from international trade journals and wire services. This dataset became the foundation for the remainder of the project.

However, collecting the dataset introduced a new set of technical challenges. TDM Studio exports its data in a complex directory structure containing thousands of

individual XML files rather than a simple spreadsheet format. To process these files I developed a custom Python pipeline using the BeautifulSoup library together with the lxml parser. The script automatically scanned each XML document and extracted the relevant textual components including the title, abstract, and full body text. These components were then combined into a single unified text field and stored within a dataframe for further analysis.

Another significant methodological challenge arose from the structure of the luxury fashion industry itself. Many of the brands that appear in media coverage are actually subsidiaries of larger corporate conglomerates. For example, brands such as Dior and Fendi are owned by the parent holding company LVMH, while Gucci and Balenciaga belong to the Kering group. This posed a major problem for the analysis because sustainability scores from rating agencies are reported only for the parent companies rather than for the individual brands. If each brand were treated as a separate entity, it would be impossible to compare the media data with corporate sustainability ratings.

To resolve this issue, I implemented a hierarchical entity resolution system that mapped more than one hundred brand names to thirty two parent companies and independent houses. This mapping ensured that whenever a news article mentioned a subsidiary brand, the reference was attributed to the correct parent company for the purposes of the analysis. For example, if an article discussed allegations of misleading sustainability claims related to Dior, the reference was counted as part of the overall dataset for LVMH. This process ensured that the media data aligned correctly with the available sustainability ratings and prevented the dataset from fragmenting into hundreds of separate brand level observations.

After constructing the dataset and resolving the entity mapping problem, the next stage of the research involved designing a quantitative metric capable of measuring greenwashing risk. Drawing on existing literature, I adopted a conceptual framework often referred to as the talk versus walk model. Within this framework, the term talk refers to public discourse about corporate sustainability initiatives, while walk refers to evidence of concrete environmental actions or policies. Greenwashing risk can therefore be conceptualized as the imbalance between these two types of communication.

To operationalize this idea, I constructed two separate textual datasets. The first dataset captured instances where news articles accused companies of greenwashing or misleading environmental claims. These articles were identified using keywords such as greenwashing, misleading claim, and deceptive marketing. The second dataset captured references to verifiable sustainability actions using terms such as climate, emissions, net zero, and circular economy. The final Greenwashing Index was calculated as the ratio between these two measures. A higher ratio indicates that accusations of misleading communication appear more frequently relative to sustainability discussions.

While this approach initially appeared straightforward, the analysis encountered a major obstacle when applying topic modeling techniques to the news corpus. The first attempt at Latent Dirichlet Allocation produced highly irrelevant topics containing words related to data centers, horse racing, and university press releases. This problem emerged because many of the downloaded documents were large daily news digests containing

multiple unrelated stories within the same file. As a result, the topic modeling algorithm identified dominant themes that were completely unrelated to sustainability.

To address this problem, the methodology was redesigned using a context window approach. Instead of feeding entire documents into the topic modeling algorithm, the Python script was modified to split each article into individual sentences. The model then analyzed only those sentences that contained explicit greenwashing related keywords. This targeted filtering process removed the vast majority of irrelevant content and forced the algorithm to focus on the specific parts of the text where sustainability criticism occurred. Additional stopwords filtering was also implemented to remove publisher boilerplate, HTML formatting tags, and commonly repeated terms that appeared across multiple documents.

Once these improvements were implemented, the topic modeling results began to reveal meaningful patterns within the dataset. The refined topics indicated that media discussions of greenwashing tend to focus on two main themes. The first theme centers on concerns about transparency and the effectiveness of corporate sustainability initiatives. The second theme highlights accusations of misleading communication directed at consumers and regulators. These themes provided valuable qualitative insight that complemented the quantitative analysis of the Greenwashing Index.

The final stage of the project involved comparing the calculated index values with corporate sustainability scores from S&P Global. Using Pearson correlation analysis, the results revealed an almost zero relationship between the two variables. In other words, companies with higher sustainability disclosure scores were not necessarily less

likely to face accusations of greenwashing in media coverage. This finding suggests that corporate sustainability ratings and public narrative operate as largely independent systems of evaluation.

Completing this project required integrating knowledge from several different areas of study. Coursework in environmental governance helped me understand the theoretical foundations of sustainability reporting and greenwashing behavior. Classes in statistics and computational methods provided the technical skills necessary to construct datasets and perform quantitative analysis. Finally, the interdisciplinary nature of the Signature Work program encouraged me to combine these perspectives into a single research framework.

Overall, the Signature Work process strengthened my ability to design independent research projects that combine theoretical insight with empirical data analysis. It also highlighted the importance of persistence and methodological flexibility when working with complex datasets. Many of the most important insights of the project emerged not from the initial research plan but from the challenges encountered along the way. Learning how to adapt to these obstacles was one of the most valuable aspects of the experience.

Looking forward, the skills developed through this project will be directly relevant to my future academic and professional goals. As sustainability reporting and environmental governance continue to evolve, organizations will increasingly require analytical tools capable of evaluating corporate claims using independent data sources. The Greenwashing Index developed in this research represents one step toward that

objective by demonstrating how computational text analysis can be used to monitor environmental narratives at scale.

APPENDICES

APPENDIX A: GDELT

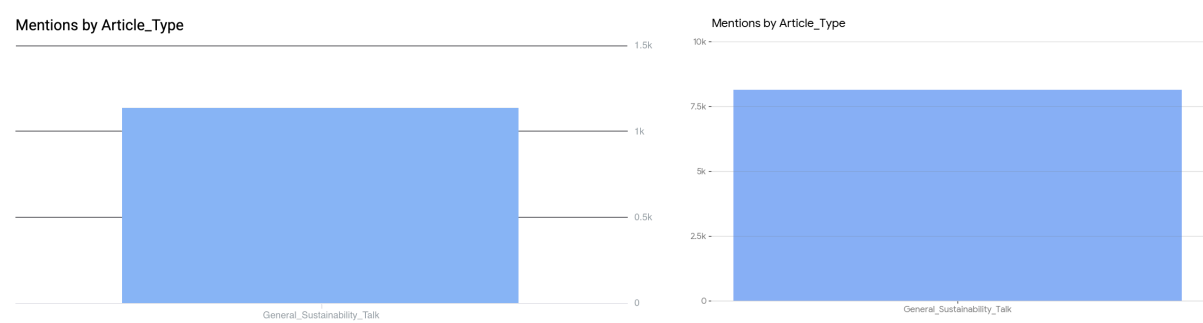


Figure 1. Mentions by Article Type (Greenwashing, Sustainability or General Sustainability Talk) for 2016-2019 (left) and 2020-2025 (right).

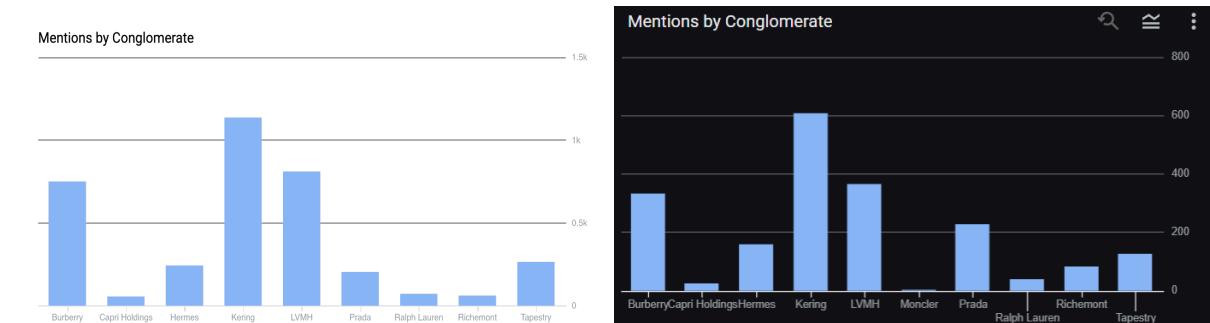


Figure 2. Mentions by Brand name for all of the 10 major luxury fashion brands for 2016-2019 (left) and 2020-2025 (right).

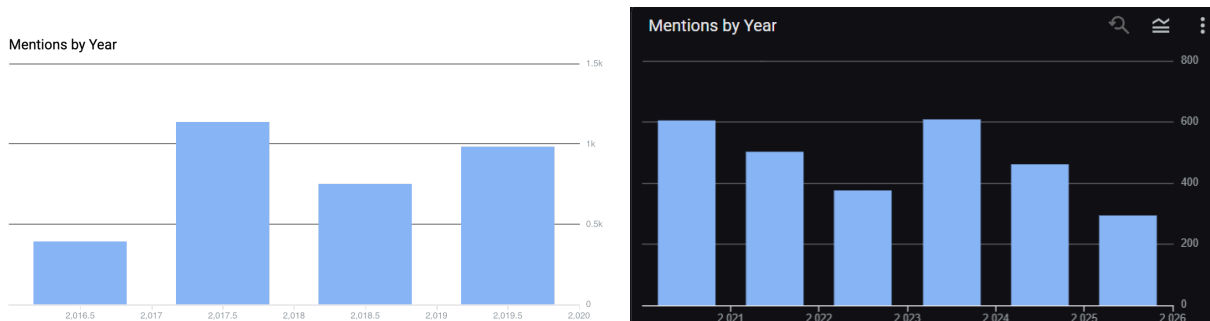


Figure 3. Mentions by year for the 10 major luxury fashion brands between the time periods of 2016-2019 (right) and 2020-2025 (left).